

Eisenstein's Criterion and the Collapsed Pandrosion Field

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Abstract

Eisenstein's irreducibility criterion states that if $p \mid a_i$ for $i < n$ and $p^2 \nmid a_0$, then P is irreducible over $\mathbb{Q}$. In Pandrosion terms: $P \equiv z^n \pmod{p}$ means $F_P \equiv n/z$, a single pole of order n at the origin. All n roots *collapse* to 0 modulo p , preventing non-trivial factorization. The Newton polygon has a single edge of slope $-1/n$, confirming irreducibility by Dumas' theorem. We verify for all cyclotomic $\Phi_p(z+1)$ (Eisenstein at p) and 100 random Eisenstein polynomials.

1 The criterion

Theorem 1.1 (Eisenstein, 1850). *Let $P(z) = z^n + a_{n-1}z^{n-1} + \dots + a_0 \in \mathbb{Z}[z]$. If there exists a prime p such that:*

1. $p \mid a_i$ for $i = 0, 1, \dots, n-1$,
2. $p^2 \nmid a_0$,

then P is irreducible over $\mathbb{Q}$.

2 Pandrosion interpretation

Proposition 2.1. *Under Eisenstein's hypotheses:*

$$\boxed{P \equiv z^n \pmod{p}, \quad F_P \equiv \frac{n}{z} \pmod{p}.} \tag{1}$$

The Pandrosion field modulo p has a single pole of order n at $z = 0$. All n roots collapse to the origin.

Proof of irreducibility. Suppose $P = QR$ over $\mathbb{Z}$. Modulo p : $z^n \equiv QR$, so $Q \equiv z^j$ and $R \equiv z^{n-j}$ for some $0 < j < n$. Then $Q(0) \equiv 0$ and $R(0) \equiv 0 \pmod{p}$, giving $a_0 = Q(0)R(0) \equiv 0 \pmod{p^2}$, contradicting hypothesis (2). $\square$

3 Newton polygon

The Newton polygon of P at p has vertices $(i, v_p(a_i))$. For Eisenstein polynomials: $(0, v_p(a_0)) = (0, 1)$ and $(n, 0)$, giving a *single edge of slope $-1/n$* . By Dumas' theorem, a single edge implies irreducibility.

P	p	Slope	Irreducible
$z^3 + 3z + 3$	3	$-1/3$	✓
$z^5 - 14z + 7$	7	$-1/5$	✓
$z^2 - 2$	2	$-1/2$	✓

4 Cyclotomic application

Corollary 4.1. *For every prime p , the cyclotomic polynomial $\Phi_p(z)$ is irreducible over $\mathbb{Q}$.*

Proof. $\Phi_p(z+1) = \frac{(z+1)^p - 1}{z} = z^{p-1} + \binom{p}{1}z^{p-2} + \cdots + \binom{p}{p-1}$. Since $p \mid \binom{p}{k}$ for $1 \leq k \leq p-1$ and $p^2 \nmid p = \binom{p}{p-1}$, Eisenstein applies. $\square$

Verified for $p = 2, 3, 5, 7, 11$: $\Phi_p(z+1)$ is Eisenstein at p .

5 Limitations

$z^4 + 1$ is irreducible over $\mathbb{Q}$ but satisfies Eisenstein at *no* prime: modulo every p , it factors non-trivially. Eisenstein is sufficient but not necessary for irreducibility.

References

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